



AWS Academy Cloud Developing
Module 01 Student Guide
Version 2.0.6

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Module 1: Welcome to AWS Academy Cloud Developing

Module overview

Sections

1. Course objectives and overview
2. Café business case introduction
3. Roles in cloud computing

Activities

AWS Documentation Scavenger Hunt



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This module includes the following sections:

1. Course objectives and overview
2. Café business case introduction
3. Roles in cloud computing

This module also includes an AWS Documentation Scavenger Hunt activity.



Section 1: Course objectives and overview.

Course prerequisites

- Completion of the **AWS Academy Cloud Foundations** course, or
- Completion of a similar level of Amazon Web Services (AWS) training or certification, such as having:
 - Passed the AWS Certified Cloud Practitioner certification exam, or
 - Completed the AWS Cloud Practitioner Essentials course, or
 - Completed the AWS Technical Essentials course
- Additionally, it is assumed that you have:
 - A working knowledge of at least one programming language supported by an AWS software development kit (SDK)
 - Experience using an SDK
 - Familiarity with using Windows or Linux, the command line, and application programming interfaces (APIs)
 - Familiarity with virtualization and distributed computing
 - An understanding of version control (for example, Git)
 - An understanding of data storage mechanisms such as SQL and NoSQL databases



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The following are the prerequisites for the AWS Academy Cloud Developing course:

- **AWS Academy Cloud Foundations:** You must have completed the AWS Academy Cloud Foundations course or have equivalent experience. Academy Cloud Foundations is an AWS Academy course that is designed to provide you with an overall understanding of the AWS Cloud, independent of specific technical roles. The course provides a detailed overview of cloud concepts, and core AWS services and their pricing, security, architecture, and support.
- **Working knowledge of at least one programming language supported by an AWS software development kit (SDK):** AWS Academy Cloud Developing requires that you have some coding experience in a language that's supported by an AWS SDK. The languages currently supported by the labs are Node.js, Ruby, and Python. For more information about the SDKs that AWS supports, see "Tools to Build on AWS" at <https://aws.amazon.com/developer/tools/>.
- Experience using an SDK
- Familiarity with using Windows or Linux, the command line, and application programming interfaces (APIs)
- Familiarity with virtualization and distributed computing
- An understanding of version control (for example, Git)
- An understanding of data storage mechanisms such as SQL and NoSQL databases

Course objectives

- Recall cloud computing services and models
- Describe developing on AWS
- Write code that interacts with Amazon Simple Storage Service (Amazon S3) by using AWS SDKs
- Explain the role of AWS Identity and Access Management (IAM)
- Write code that interacts with Amazon DynamoDB by using AWS SDKs
- Create a REST API by using Amazon API Gateway
- Create functions with AWS Lambda by using AWS SDKs
- Configure containers
- Explain caching with Amazon CloudFront and Amazon ElastiCache
- Develop solutions with Amazon Simple Queue Service (Amazon SQS) and Amazon Simple Notification Service (Amazon SNS)
- Describe the use of AWS Step Functions
- Explain how to build secure applications
- Identify best practices for deploying applications



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After completing this course, you will be able to do the following:

- Recall cloud computing services and models
- Describe developing on Amazon Web Services (AWS)
- Write code that interacts with Amazon Simple Storage Service (Amazon S3) by using AWS SDKs
- Explain the role of AWS Identity and Access Management (IAM)
- Write code that interacts with Amazon DynamoDB by using AWS SDKs
- Create a REST API by using Amazon API Gateway
- Create functions with AWS Lambda by using AWS SDKs
- Configure containers
- Explain caching with Amazon CloudFront and Amazon ElastiCache
- Develop solutions with Amazon Simple Queue Service (Amazon SQS) and Amazon Simple Notification Service (Amazon SNS)
- Describe the use of AWS Step Functions
- Explain how to build secure applications
- Identify best practices for deploying applications

Course outline

- Module 1: Welcome to AWS Academy Cloud Developing (this module)
- Module 2: Introduction to Developing on AWS
- Module 3: Developing Storage Solutions
- Module 4: Securing Access to Cloud Resources
- Module 5: Developing Flexible NoSQL Solutions
- Module 6: Developing REST APIs
- Module 7: Developing Event-Driven Serverless Solutions
- Module 8: Introducing Containers and Container Services
- Module 9: Caching Information for Scalability
- Module 10: Developing with Messaging Services
- Module 11: Defining Workflows to Orchestrate Functions
- Module 12: Developing Secure Applications on AWS
- Module 13: Automating Deployment Using CI/CD Pipelines
- Module 14: Bridging to Certification



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To achieve the course objectives, the course consists of the 14 modules that are listed on the slide. The next slides provide more detail on what subtopics each module covers.

Module 2: Introduction to Developing on AWS

Module sections

1. Introduction
2. Systems development lifecycle
3. Steps to get started developing on AWS
4. Fundamentals of working with the AWS SDKs

Module lab

Exploring AWS CloudShell and an IDE



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Module 2 covers the following topics:

1. Introduction
2. Systems development lifecycle
3. Steps to get started developing on AWS
4. Fundamentals of working with the AWS SDKs

At the end of this module, you will complete the Exploring AWS CloudShell and an IDE lab.

Module 3: Developing Storage Solutions

Module sections

1. Introduction
2. Introducing Amazon S3
3. Creating S3 buckets
4. Working with S3 objects
5. Protecting data and managing access to Amazon S3 resources



Module lab

Working with Amazon S3



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Module 3 covers the following topics:

1. Introduction
2. Introducing Amazon S3
3. Creating S3 buckets
4. Working with S3 objects
5. Protecting data and managing access to Amazon S3 resources

At the end of this module, you will complete the Working with Amazon S3 lab.

Module 4: Securing Access to Cloud Resources

Module sections

1. Introduction
2. Shared responsibility model
3. Introducing AWS Identity and Access Management (IAM)
4. Authenticating with IAM
5. Authorizing with IAM



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Module 4 covers the following topics:

1. Introduction
2. Shared responsibility model
3. Introducing AWS Identity and Access Management (IAM)
4. Authenticating with IAM
5. Authorizing with IAM

Module 5: Developing Flexible NoSQL Solutions

Module sections

1. Introduction
2. Introducing AWS database options
3. Key concepts for DynamoDB
4. Partitions and data distribution
5. Secondary indexes
6. Read/write throughput
7. Streams and global tables
8. Backup and restore
9. Basic operations for DynamoDB tables



Module lab

Working with DynamoDB



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Module 5 covers the following topics:

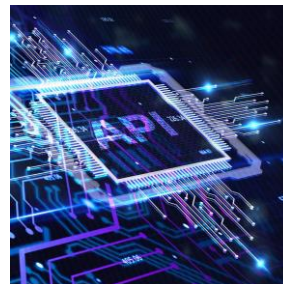
1. Introduction
2. Introducing AWS database options
3. Key concepts for DynamoDB
4. Partitions and data distribution
5. Secondary indexes
6. Read/write throughput
7. Streams and global tables
8. Backup and restore
9. Basic operations for DynamoDB tables

At the end of this module, you will complete the Working with DynamoDB lab.

Module 6: Developing REST APIs

Module sections

1. Introduction
2. Introducing APIs
3. Introducing Amazon API Gateway
4. Creating a REST API
5. Integrating with API Gateway
6. Deploying an API
7. Controlling access to a REST API
8. Monitoring a REST API
9. Optimizing API Gateway



Module lab

Developing REST APIs with API Gateway



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Module 6 covers the following topics:

1. Introduction
2. Introducing APIs
3. Introducing Amazon API Gateway
4. Creating a REST API
5. Integrating with API Gateway
6. Deploying an API
7. Controlling access to a REST API
8. Monitoring a REST API
9. Optimizing API Gateway

At the end of this module, you will complete the Developing REST APIs with API Gateway lab.

Module 7: Developing Event-Driven Serverless Solutions

Module sections

1. Introduction
2. Introducing serverless computing
3. Introducing AWS Lambda
4. Invoking Lambda functions
5. Setting permissions for Lambda
6. Authoring and configuring Lambda functions
7. Deploying Lambda functions
8. Monitoring and debugging tools for application developers



Module lab

Creating Lambda Functions Using the AWS SDK for Python



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Module 7 covers the following topics:

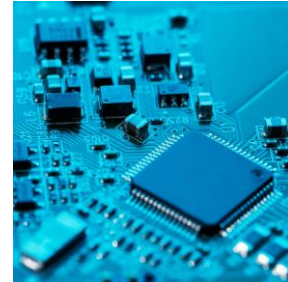
1. Introduction
2. Introducing serverless computing
3. Introducing AWS Lambda
4. Invoking Lambda functions
5. Setting permissions for Lambda
6. Authoring and configuring Lambda functions
7. Deploying Lambda functions
8. Monitoring and debugging tools for application developers

At the end of this module, you will complete the Creating Lambda Functions Using the AWS SDK for Python lab.

Module 8: Introducing Containers and Container Services

Module sections

1. Introduction
2. Introducing containers
3. Introducing Docker containers
4. Using containers for microservices
5. Introducing AWS container services
6. Deploying applications with Elastic Beanstalk



Module labs

- Migrating a Web Application to Docker Containers
- Running Containers on a Managed Service



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Module 8 covers the following topics:

1. Introduction
2. Introducing containers
3. Introducing Docker containers
4. Using containers for microservices
5. Introducing AWS container services
6. Deploying applications with AWS Elastic Beanstalk

At the end of this module, you will complete two labs: Migrating a Web Application to Docker Containers and Running Containers on a Managed Service.

Module 9: Caching Information for Scalability

Module sections

1. Introduction
2. Caching overview
3. Caching with ElastiCache
4. Caching with CloudFront
5. Caching strategies

Module labs

- Caching Application Data with ElastiCache
- Implementing CloudFront for Caching and Application Security



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Module 9 covers the following topics:

1. Introduction
2. Caching overview
3. Caching with ElastiCache
4. Caching with CloudFront
5. Caching strategies

At the end of this module, you will complete two labs: Caching Application Data with ElastiCache and Caching with CloudFront.

Module 10: Developing with Messaging Services

Module sections

1. Introduction
2. Processing requests asynchronously
3. Introducing Amazon SQS
4. Working with Amazon SQS messages
5. Configuring Amazon SQS queues
6. Introducing Amazon SNS
7. Developing with Amazon SNS
8. Introducing Kinesis Data Streams



Module lab

Implementing a Messaging System Using Amazon SNS and Amazon SQS



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Module 10 covers the following topics:

1. Introduction
2. Processing requests asynchronously
3. Introducing Amazon SQS
4. Working with Amazon SQS messages
5. Configuring Amazon SQS queues
6. Introducing Amazon SNS
7. Developing with Amazon SNS
8. Introducing Amazon Kinesis Data Streams

At the end of this module, you will complete the Implementing a Decoupled Architecture Using Amazon SNS and Amazon SQS lab.

Module 11: Defining Workflows to Orchestrate Functions

Module sections

1. Introduction
2. Coordinating tasks in distributed applications
3. Introducing Step Functions
4. State types
5. Step Functions use cases
6. Step Functions API



Module lab

Orchestrating Serverless Functions with Step Functions



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Module 11 covers the following topics:

1. Introduction
2. Coordinating tasks in distributed applications
3. Introducing Step Functions
4. State types
5. Step Functions use cases
6. Step Functions API

At the end of this module, you will complete the Orchestrating Serverless Functions with Step Functions lab.

Module 12: Developing Secure Applications on AWS

Module sections

1. Introduction
2. Securing network connections
3. Authenticating with AWS STS
4. Authenticating with Amazon Cognito

Module lab

Implementing Application Authentication Using Amazon Cognito



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Module 12 covers the following topics:

1. Introduction
2. Securing network connections
3. Authenticating with AWS Security Token Service (AWS STS)
4. Authenticating with Amazon Cognito

At the end of this module, you will complete the Implementing Application Authentication Using Amazon Cognito lab.

Module 13: Automating Deployment Using CI/CD Pipelines

Module sections

1. Introduction
2. Introducing DevOps
3. Using AWS code services for CI/CD
4. Deploying applications with CloudFormation
5. Deploying applications with AWS SAM



Module lab

Automating Application Deployment Using a CI/CD Pipeline



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Module 13 covers the following topics:

1. Introduction
2. Introducing DevOps
3. Using AWS code services for continuous integration and continuous deployment (CI/CD)
4. Deploying applications with AWS CloudFormation
5. Deploying applications with the AWS Serverless Application Model (AWS SAM)

At the end of this module, you will complete the Automating Application Deployment Using a CI/CD Pipeline lab.

Module 14: Bridging to Certification

Module sections

- Certification and resources
- Additional resources



aws
certified



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Module 14 focuses on bridging your learning in this course to attaining an AWS certification. Module 14 covers the following topics:

- Certification and resources
- Additional resources



Section 2: Café business case introduction.

Café and bakery



The labs in this course are built around a fictional business case. The business case provides a way to explore cloud-computing topics in the context of relatable business needs. This scenario is intended to provide an example of the real-world applicability of the technical concepts that you will learn in this course.

Café owners and staff

Frank

- Co-owner of café
- Retired from the Navy
- Likes to bake
- Not technical



Martha

- Co-owner of café
- Retired accountant
- Knows how to use spreadsheets, otherwise not technical



Sofía

- Daughter of Frank and Martha
- Manages the café's supply chain
- Technical skills, including programming, future business administration student
- Started to use AWS



Nikhil

- Café employee, visual design skills
- Interested in learning cloud computing
- Might take on more responsibilities at the café when Sofía starts her studies at the university



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Frank and Martha had a dream to open a café and bakery after they retired. They were not yet ready to live out their retirement around the house. Instead, Frank and Martha wanted to do something that included their love of baking and supplemented their income. They enjoy interacting with the people in their neighborhood. They also like to support community events across town with their baked goods and coffee.

To make their dreams a reality, Frank and Martha recently opened their café and bakery. Their daughter, Sofía, and a local school student, Nikhil, help out and work at the café. Since they opened the café, Frank and Martha have experienced an increase in local business. They also sometimes have customers who are traveling through the area either for business or as tourists.

Sofía is Frank and Martha's daughter, and she runs the café. She manages the supply chain to source ingredients, and she tracks inventory. She took some programming classes in secondary school and plans to start working toward a university degree in business administration later this year.

Sofía just learned about Amazon Web Services. She has talked to her parents about how they could use AWS services to automate some aspects of the café business, reduce manual administrative work, and improve the customer experience.

Nikhil works at the café part time. He will finish secondary school this year. He works behind the counter to serve customers and do other tasks under Sofía's supervision. He has some experience with visual design, and wants to learn more about web development and cloud computing. He plans to get a university degree that will build upon his existing design skills and also enable him to learn cloud computing skills.

AWS consultants, café visitors

Olivia

- An AWS solutions architect
- Technical, with specialties in databases and network technologies



Mateo

- Systems administrator and engineer
- Likes to find ways to automate and create repeatable solutions
- Knows the importance of backups and disaster recovery in solution design



Faythe

- Developer, experienced with AWS programming interfaces
- Knowledgeable about cloud security



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Olivia is an AWS solutions architect (SA) who recently moved to the downtown area. She frequently visits the café and enjoys talking with Sofía. Through their conversations, Sofía has learned that Olivia is an expert in AWS and cloud technologies. Olivia used to be a network engineer, and she also has a strong background in database technologies.

Faythe is an AWS developer who recently completed an AWS internship program. She likes to use her programming skills to apply the appropriate technology to a business problem. She recently achieved the AWS Certified Security – Specialty and is interested in developing data analytics solutions. Faythe is friends with Olivia and Mateo, and she also frequently visits the café.

Mateo is an experienced AWS SysOps engineer. He is skilled at bringing automation and fault tolerance to the solutions that he builds. He also likes to design for backup and disaster recovery. Mateo previously worked as a developer and has been a mentor to Faythe since she started as an intern with AWS. Mateo is happy to help Faythe and anyone who is interested in learning. Mateo is friends with Nikhil, Sofía, Olivia, and Faythe, and he frequently grabs a coffee at the café on his way to work.

You will build solutions for the café in the labs in this course

The café has business needs that can be solved with cloud development.



The café employees and the consultants often socialize and share cloud architecture ideas.



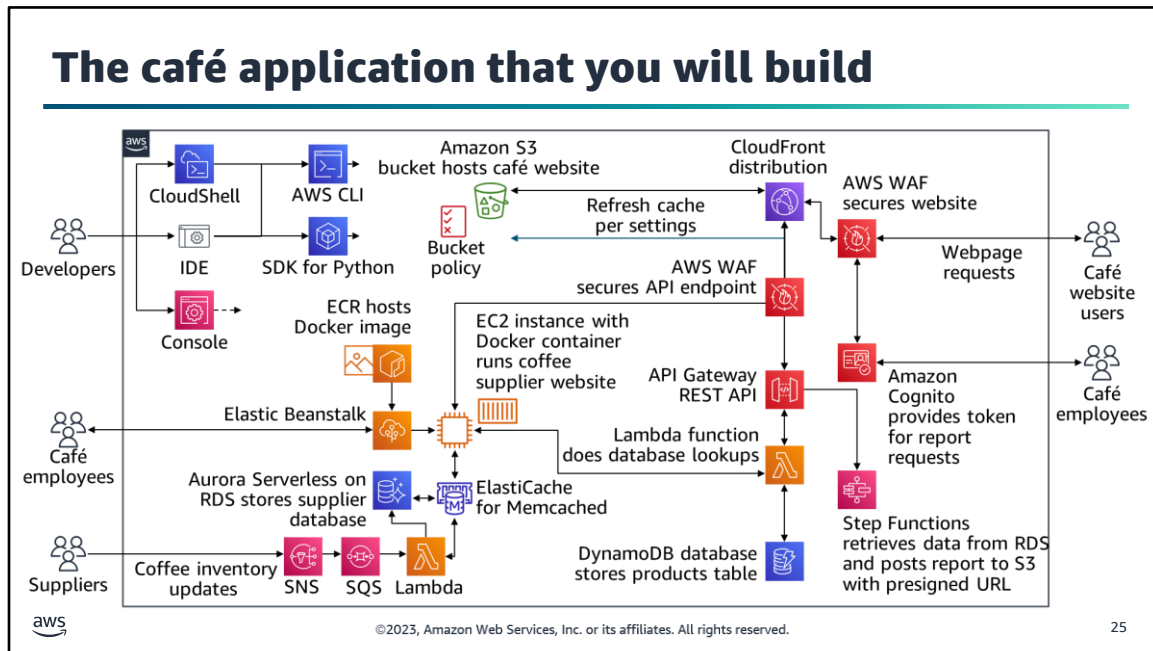
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In the hands-on labs in this course, you gain hands-on practice building on AWS.

Each lab presents new business requirements, which are based on the evolving needs of the café. Some labs contain sections where the instructions do not provide full, click-level guidance or detailed step-by-step instructions. Rather, you will be challenged to apply the skills that you gained from previous labs and the concepts that are presented in the lectures.

In the labs, you take on the role of Sofía or Nikhil. With advice from the AWS consultants who occasionally pass through the café, you will architect cloud solutions that help fulfill the business needs of the café.



The diagram on this slide gives an overview of the application that you will build through the labs in this course. By the end of this course, you will have learned about all the components in this diagram. You should also be able to construct your own cloud application that has as many components as this example.

Here is an overview of the services and application logic represented in the diagram: Developers use cloud development tools, including the AWS Management Console, AWS Command Line Interface (AWS CLI), AWS CloudShell, and the AWS SDK for Python (Boto3) to interact with and build the application.

- The café website has static content stored in the Amazon Simple Storage Service (Amazon S3). The S3 bucket is secured with a bucket policy.
- Dynamic product information, which is accessible through the website, is stored on Amazon DynamoDB in the Products table.
- Café website user requests are routed through AWS WAF and Amazon CloudFront to retrieve content from the S3 bucket, an Amazon API Gateway endpoint, or both. The API endpoints are integrated with an AWS Lambda function, which handles queries to the Products table in DynamoDB.
- The café website also integrates data from a legacy coffee suppliers web application, which was originally built by a company that the café acquired. The application was subsequently migrated to run in containers on AWS Elastic Beanstalk. Docker containers running on Amazon Elastic Compute Cloud (Amazon EC2) instances host the front end of the suppliers website. The container image is pulled from Amazon Elastic Container Registry (Amazon ECR).
- The suppliers website backend was migrated from MySQL to instead run on an Amazon Aurora Serverless database on Amazon Relational Database Service (Amazon RDS) with Amazon ElastiCache for Memcached providing a database cache.
- Messaging services including Amazon Simple Notification Service (Amazon SNS) and Amazon Simple Queue Service (Amazon SQS) are used to asynchronously

manage the flow of updates from suppliers about their coffee inventories.

- Café employees can request a coffee inventory report, and AWS Step Functions coordinates the workflow to fulfill report requests.
- Amazon Cognito user pools control access to portions of the café website that are not available for all users, such as the inventory report.

You will see this diagram repeated at the start of most modules in the course. Components of the entire application will be revealed in the diagram within the modules where you will first implement or use those components in a lab activity.



Section 3: Roles in cloud computing.

Roles in cloud computing: IT professional



Characteristics

- Generalist, might manage an application
- Often manages a production environment
- Highly technical
- Might have significant or limited experience in cloud technologies
- Might specialize in one area (such as security or storage)

Job titles: IT administrator, systems administrator, network administrator



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In this section, you will learn about five common roles in cloud computing.

You might want to start your career in cloud computing or to transition your career to a cloud computing role. Maybe you would like to work in an organization where some employees have cloud computing responsibilities. For any of these reasons, it is helpful to understand the common job titles or roles that individuals, teams, or departments perform.

IT professionals are generalists. They typically have a broad range of skills. For example, they might manage the infrastructure for an entire application and have a strong understanding of the components that make up the solution. However, they might not always have detailed knowledge of any one service that is part of the application. IT professionals are typically highly technical.

Common job titles include IT administrator, systems administrator, and network administrator.

Roles in cloud computing: IT leader



Characteristics

- Leads a team of IT professionals
- Responsible for day-to-day operations
- Manages a budget, and stays informed about and chooses new technologies
- Hands on during early stages of a project, then delegates the team to take over

Job titles: IT manager, IT director, IT supervisor



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IT leaders are managers. They typically lead a team of IT professionals and decide on the type of technology that will be used for a project. They might be significantly involved in the implementation details early in the project lifecycle. Then, they delegate the team to handle the details as the project gets closer to completion.

Typical job titles include IT manager, IT director, and IT supervisor.

Roles in cloud computing: DevOps engineer



Characteristics

- Builds out the infrastructure that applications run on, often in the cloud
- Follows the guidelines of the cloud architect
- Prefers experimenting and trying things out rather than a lot of reading

Job titles: DevOps engineer, reliability engineer, build engineer



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DevOps engineers spend their time building out the infrastructure that applications run on. They often create or improve the code that installs and configures servers and application deployments. DevOps staff prefer to experiment and learn by doing. They create repeatable deployment solutions, and they work to apply engineering skills to the business needs of operations teams.

Common job titles include DevOps engineer, reliability engineer, and build engineer.

Roles in cloud computing: Cloud architect



Characteristics

- Stays up to date with new technologies and helps decide which to use
- Provides documentation, processes, and tooling to developers
- Gives developers freedom to innovate
- Common challenges include:
 - Resource management
 - Cost optimization
 - Defining best practices for performance, reliability, and security

Job titles: cloud architect, systems engineer, systems analyst



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Cloud architects spend their time reading and staying up to date with the latest developments and trends in cloud computing. They are responsible for the design architecture of applications and selecting which technologies should be used to meet the needs of a technical business objective. They should be aware of the capabilities of the many cloud service options that are available. Thus, they can decide which services to adopt, given a specific set of business requirements. Cloud architects provide guidance to developers through architectural diagrams and documentation. They also provide tooling, but they give the development team room to innovate if they meet the success criteria.

Common challenges for the cloud architect role include resource management; cost optimization; and defining best practices for performance, reliability, and security.

Typical job titles include cloud architect, systems engineer, and systems analyst.

Roles in cloud computing: Developer



Characteristics

- Writes, tests, and fixes code
- Thinks about projects at the application level
- Likes sample code
- Works with APIs and SDKs

Job titles: software developer, systems architect, software development manager

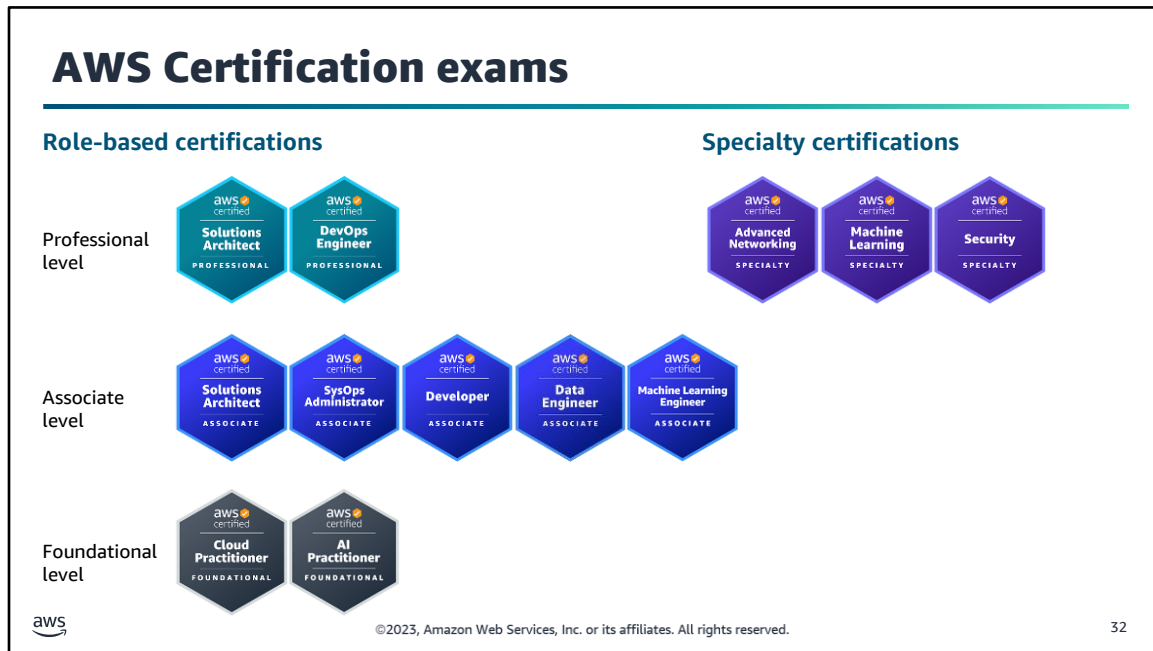


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Another common role in cloud computing is the developer role. *Developers* like to code. They work with the details—writing, testing, and fixing the code that makes an application work. Developers are likely to borrow ideas from sample code. They work with application programming interfaces (APIs) and software development kits (SDKs).

Common job titles include software developer, systems architect, and software development manager.



AWS Certification helps learners to build credibility and confidence by validating their cloud expertise with an industry-recognized credential. Certification helps organizations to identify skilled professionals who can lead cloud initiatives by using AWS.

The slide shows the AWS Certifications that are currently available. To earn an AWS Certification, you must earn a passing score on a proctored exam. Each certification level for role-based certifications provides a recommended experience level with AWS Cloud services as follows:

- Professional: two years of comprehensive experience designing, operating, and troubleshooting solutions using the AWS Cloud
- Associate: One year of experience solving problems and implementing solutions using the AWS Cloud
- Foundational: Six months of fundamental AWS Cloud and industry knowledge

This course helps prepare you for the AWS Certified Developer – Associate exam

Specialty certifications focus on a particular technical domain. The recommended experience for taking a specialty exam is technical experience in the domain as specified in the exam guide.

AWS does not publish a list of all services or features that are covered in a certification exam. However, the exam guide for each exam lists the current topic areas and objectives that the exam covers. For exam guides and other preparation materials, see “AWS Certification exam preparation” at <https://aws.amazon.com/certification/certification-prep/>.

The information on this slide is current as of October, 2024. However, exams are frequently updated, and the details regarding which exams are available—and what is tested by each exam—are subject to change. For the latest AWS Certification exam information, see “AWS Certification” at <https://aws.amazon.com/certification/>.

You are required to update your certification (or recertify) every 3 years. For more information on recertification, see “AWS Recertification” at <https://aws.amazon.com/certification/recertification/>.

The final module in this course provides more information to supplement what you learn in this course. That module will also guide you in how to apply this knowledge toward achieving the AWS Certified Developer – Associate certification.

Activity: AWS Documentation Scavenger Hunt



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You will now complete the AWS Documentation Scavenger Hunt.

The purpose of this brief activity is to acclimate you to locating information in the AWS Documentation, which will support your learning experience throughout the course.

We recommend bookmarking the AWS Documentation site for future use:

<https://docs.aws.amazon.com>.

AWS Documentation Scavenger Hunt

1. Which section of the website contains links to the supported AWS SDKs and toolkits?
2. How would you navigate to the method for creating an S3 bucket in the AWS SDK for Python (Boto3)?
3. How would you navigate to the Getting Started topic in the AWS Documentation for IAM?
4. Which section of the website has a tutorial on how to build a serverless application?
5. Which section of the website provides a link to AWS technical whitepapers?

<https://docs.aws.amazon.com>



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Use the AWS Documentation website at <https://docs.aws.amazon.com> to answer the following questions:

1. Which section of the website contains links to the supported AWS SDKs and toolkits?
2. How would you navigate to the method for creating an S3 bucket in the AWS SDK for Python (Boto3)?
3. How would you navigate to the Getting Started topic in the AWS Documentation for IAM?
4. Which section of the website has a tutorial on how to build a serverless application?
5. Which section of the website provides a link to AWS technical whitepapers?



It's now time to review the module.

Module summary

In summary, in this module, you learned to:

- Identify course prerequisites and objectives
- Recognize the café business case
- Identify roles in cloud computing
- Familiarize yourself with AWS documentation

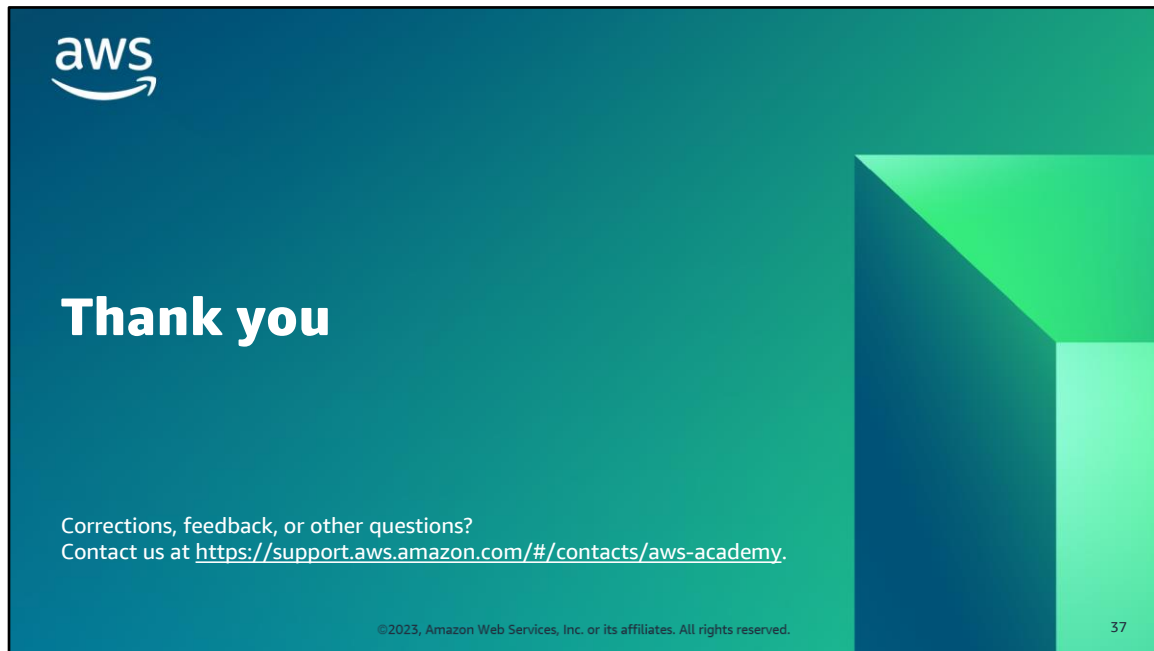


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In summary, in this module, you learned how to do the following:

- Identify course prerequisites and objectives
- Recognize the café business case
- Identify roles in cloud computing
- Familiarize yourself with AWS documentation



Thank you for completing this module.